

Resource Trading Incentive Mechanism Based on Stackelberg Game and Auction Theory in Computing Power Network

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Abstract—Computing power network (CPN) is a new architecture proposed in recent years. It has integrated multi-dimensional resources, such as computing, network and storage resources in cloud, edge and end to realize more flexible resource allocation and task scheduling. Consumers in CPN want to purchase some resources to meet their service demands and gain some profits, while operators would like to obtain payments by selling their idle resources to others. However, consumers usually prefer to purchase resources from large and public cloud operators in reality, while small and private operators who lack the recognition struggle to obtain the opportunity to join in the sales. Therefore, both parties may lack the passion and enthusiasm to participate in CPN transaction and there is a great need for a resource trading incentive mechanism. In this paper, we propose a three-tier joint optimization framework to increase their motivations and establish a centralized computing service platform (CSP). We use the Stackelberg game to describe the resource-purchasing process of computing service demanders (CSDs) and use the auction theory to describe the task-scheduling process of computing service operators (CSOs). In this way, both parties can gain their profits so that they will have a greater willingness to participate in the transaction. The simulation results demonstrate the feasibility and superiority of our scheme.

Keywords—computing power network, resource transaction, Stackelberg game, auction theory

I. INTRODUCTION

In the future era of intelligent connectivity between humans, objects and everything, compute-intensive and delay-sensitive tasks have made higher requirements for faster network connectivity, on-demand resource usage, and flexible service orchestration. Computing power network (CPN) is a new architecture proposed in recent years and the integration of computing and network will become a significant trend in the evolution of network [1]. CPN resources have the universal characteristics which are similar to electricity. We can usually establish a unified resource transaction platform like the electric operators to collect and schedule idle resources in the network. Operators with idle resources can obtain some profits through sales and consumers in the network can purchase resources from the platform anytime and anywhere to

meet their service demands, without caring for who provides these resources or where they are deployed [2]. Therefore, CPN can provide a tight connection between the operators and consumers and arranges multi-dimensional resources in a more refined manner within the network.

The resource transaction between consumers and operators is an important issue in CPN scenarios. Blockchain has been widely adopted to assist in transaction [3] [4] while the suitable trading incentive scheme is still being explored. In traditional scenarios, such as mobile edge computing (MEC), internet of vehicles (IoV) and blockchain, game theory has been widely used to solve the resource allocation and task scheduling problems. Authors in [5] summarized the applications of game theories in edge computing systems. A distributed MEC scheme based on the Stackelberg game was designed in [6], and the unified and differential pricing algorithms were proposed correspondingly. A two-stage Stackelberg game scheme was developed in [7], and discussed the pricing interaction between service operators and blockchain miners to solve the problem of resource management in edge computing. Authors in [8] proposed a reliable combinatorial bidirectional auction mechanism and the dynamic pricing strategy in resource-constrained MEC environments. To attract more mobile devices to participate in task executions, the authors in [9] designed an auction scheme between task owners and mobile devices. However, these studies only discuss the situation that there is only one-dimensional resource in the network.

The types of resources in reality may be more complex, including multiple dimensions such as computing, networking, storage, and even intelligence. The consideration of resources in the network is gradually shifting from single computing resources to heterogeneous and multidimensional ones. In [10], authors proposed a combinatorial bidirectional auction mechanism for the transaction involving multiple composite resources, which improved the success rate of transaction matching and simplified the complexity of resources matching. Authors in [11] studied the allocation and pricing of multiple resources in the MEC scenario. However, the situation may also be more complex in reality, and there may be more than two participants in the transaction. In [12], authors designed a three-stage Stackelberg game process to study the optimization goals of resource allocation, content pricing, and

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quality of service. In the fog computing scenario, a three-tier model was established for data service subscribers, data service operators and fog nodes in [13] [14], and a joint optimization framework was proposed based on Stackelberg game and matching algorithm.

Although there have been many studies discussing the problem in the transaction of multiple participants involving multidimensional resources, there are still some challenges for CPN transaction. The reasons are as follows: 1) Consumers are more willing to purchase resources from large and public operators, and the market monopoly makes the smaller ones more difficult to sell resources. 2) Large operators in the network usually have their own transaction pricing strategies, and the smaller ones usually have difficulties in looking for trading partners. 3) Resources are often sold in fixed configurations and fixed prices so that the operators and consumers cannot conduct pricing negotiations subjectively. 4) Consumers only want to purchase a certain amount of resources to meet their demands and don't want to pay extra attention to the geographical location or other factors of resources.

To solve these problems, we propose a three-tier joint optimization framework and establish a CPN trading platform. By providing an appropriate transaction incentive mechanism, specifically, a more flexible resource allocation and pricing strategy, our mechanism can encourage the two parties to participate in CPN transaction. The main contributions of this paper are as follows:

- We propose a three-tier architecture to describe the process of transaction among the computing service operators (CSOs), computing service platform (CSP), and computing service demanders (CSDs).
- We discuss the demands matching interaction process between the CSDs and CSP based on the Stackelberg game and derive the equilibrium solution.
- We provide the resources matching transaction problem between the CSOs and CSP based on auction theory and demonstrate the superiority of our scheme.

The rest of the paper is organized as follows. First, we introduce the architecture of our system, and provide the models of CPN demands and resources in Section II. Section III provides the transaction interaction process and optimization problems in two stages. In Section IV, we discuss the simulation results. And we summarize the work of the paper in Section V.

II. SYSTEM DESCRIPTION

In this section, we will propose the architecture of our system firstly, including multiple CSOs, single CSP, and multiple CSDs. Then the demands model of CSDs and the resources model of CSOs will be introduced.

A. System Architecture

As illustrated in Fig.1, we consider a three-tier CPN trading architecture consisting of CSOs, CSP and CSDs. CSOs register and sell their resources to CSP, and CSP serves as an intermediary to sell resources to CSDs. Assuming there are M CSDs in the network, labeled as $D = \{d_1, d_2, \dots, d_M\}$, and N CSOs

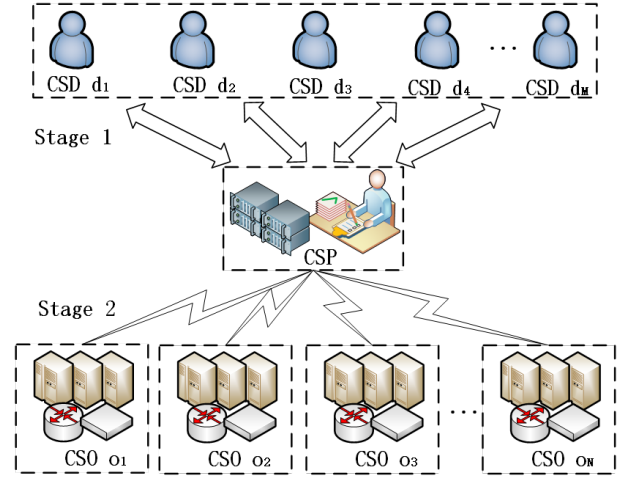


Fig. 1. System Architecture

labeled as $O = \{o_1, o_2, \dots, o_N\}$. And there is one CSP in the network as the intermediary between the two parties, which may be a distributed platform system. In this paper, we regard CSP as one single node for simplification. CSDs and CSOs register their respective qualifications to CSP, including the configurations, geographical locations, and regional electricity bills. We define $\Gamma_{pd} = \{\tau_1^d, \tau_2^d, \dots, \tau_M^d\}, \forall \tau_i^d \in \{0, 1\}$ as the service list of CSDs, and $\Gamma_{po} = \{\tau_1^o, \tau_2^o, \dots, \tau_N^o\}, \forall \tau_i^o \in \{0, 1\}$ as the service list of CSOs. Each item of these represents whether CSD d_i or CSO o_j participate in the transaction.

B. Demands Model

As for the task $X_i = \{x_{i1}, x_{i2}, x_{i3}\}$, in which x_{i1}, x_{i2}, x_{i3} represent the computing, network, and storage demands separately, we can get $C_i = (f_1(x_{i1}), f_2(x_{i2}), f_3(x_{i3}))$. Each item of it represent the mapping to the computing, network, and storage resources. We can use C_i as the comprehensive resource measurement mapping of CSD $d_i, \forall d_i \in D$ corresponding to the demands, which can be represented as

$$C_i = \sum_{k=1}^3 c_{ik} = \omega_1 f_1(x_{i1}) + \omega_2 f_2(x_{i2}) + \omega_3 f_3(x_{i3}) \quad (1)$$

ω_1, ω_2 , and ω_3 are mapping proportion factors corresponding to computing, network, and storage demands. We select the tolerable delay as the core indicator to model the service demands, and the task of CSD $d_i, \forall d_i \in D$ can be defined as $\{C_i, T_i, P_i^{max}\}$, in which T_i is the maximum tolerable delay, and P_i^{max} is the maximum price expenditure that can be accepted by CSD d_i . To facilitate measurement and expression, the concept of average price [10] can be defined as

$$p_i^{max} = \frac{P_i^{max}}{C_i} = \frac{P_i^{max}}{\omega_1 f_1(x_{i1}) + \omega_2 f_2(x_{i2}) + \omega_3 f_3(x_{i3})} \quad (2)$$

C. Resources Model

As for the resource $Q_j = (q_{j1}, q_{j2}, q_{j3})$, in which q_{j1}, q_{j2}, q_{j3} represent the computing, network, and storage resources

separately, we can comprehensively measure it as

$$Q_j = \sum_{k=1}^3 q_{jk} = \omega_1 q_{j1} + \omega_2 q_{j2} + \omega_3 q_{j3} \quad (3)$$

The resources of CSO $o_j, \forall o_j \in O$ can be defined as $\{Q_j, P_j^{\min}\}$, in which $P_j^{\min}$ is minimum acceptable price payment of CSO o_j . We define its average price as

$$p_j^{\min} = \frac{P_j^{\min}}{Q_j} = \frac{P_j^{\min}}{\omega_1 q_{j1} + \omega_2 q_{j2} + \omega_3 q_{j3}} \quad (4)$$

III. TWO-STAGE TRANSACTION INCENTIVE MECHANISM

In this section, we describe the workflow of our system firstly and divide the transaction into two stages. Then the optimization problems in each stage are introduced, and the utilities of three parties are formulated.

A. System Workflow

CSOs and CSDs are encouraged to participate in the transaction through pricing games. CSDs will get better user experience from more flexible resource allocation and billing methods, and more small and private operators will have the opportunity to sell their idle resources. At the same time, CSP can also get corresponding profits and increase its influence.

- **CSP-CSDs interaction stage:** 1) CSP obtains the service demands of CSDs and provides an initial unit price. 2) CSDs comprehensively consider their demands and CSP's price to determine their purchases. 3) CSP adjusts and updates the price based on the actual purchases.
- **CSP-CSOs interaction stage:** 1) CSP measures the resources and calculates the average prices of CSOs. 2) Based on the known quantities and prices of resources, CSP will match and select the appropriate CSOs for the transaction.

B. Trading Stage Based on Stackelberg Game

We discuss the trading process between CSDs and CSP in this section as a Stackelberg game transaction issue. Firstly, we model the utilities of both parties and formulate the optimization problem. Then we derive the equilibrium solution of the Stackelberg game problem.

1) **CSDs' Utility Model:** The benefits of CSDs usually come from the tasks or brought by the reduction of delay. At the same time, the cost usually includes resource expenses and offloading energy consumption. Therefore, we can define the utility function of CSDs as

$$U_d = \sum_{i=1}^M \tau_i^d [\alpha_i F(C_i) + \beta_i (T_i - t_i) - \gamma_i p_i q_i - \delta_i E_i] \quad (5)$$

$F(C_i)$ is the benefit mapping function and is usually monotonically increasing. t_i is the actual delay of the task. p_i is the unit price and q_i is the comprehensive measurement of purchased resources. E_i is the energy consumption and only the energy consumption of task offloading will be included [15]. $\alpha_i, \beta_i, \gamma_i$ and δ_i are the mapping factors. The delay

often includes two parts: offloading transmission delay and task processing delay, which is defined as

$$t_i = T_i^p + \frac{C_i}{q_i} \quad (6)$$

T_i^p is the maximum transmission delay guaranteed by CSP, which is determined by the distance and channel parameters. The offloading energy consumption can be defined as

$$E_i = \sum_{j=1}^N \tau_{ij}^o e_{ij} \quad (7)$$

τ_{ij}^o represents whether CSD $d_i, \forall d_i \in D$ matched with CSO $o_j, \forall o_j \in O$ or not, and e_{ij} represents the offloading energy consumption of CSD d_i , usually related to the network link and transmission power.

2) **CSP and CSOs' Joint Utility Model:** In this part, we temporarily consider CSP and CSOs as a whole. The main source of income comes from the payment of CSDs, while the main expenditure is the energy consumption and resource maintenance fees. So the joint utility can be defined as

$$\Pi_p = \sum_{i=1}^M \tau_i^d p_i q_i - \mu E_R - \sum_{i=1}^M \tau_i^d \eta q_i \quad (8)$$

μ is the energy consumption mapping factor. E_R represents the total energy consumption of resources. η is the unit maintenance fee of resources. The energy expenditure is usually related to the actual demands and resources of CSD d_i [16]. We use ρ as the energy consumption mapping factor, so the energy consumption can be expanded as

$$E_R = \sum_{i=1}^M \tau_i^d \rho C_i (q_i)^2 \quad (9)$$

3) **Problem Formulation:** There exists an one-to-many Stackelberg game that CSP serves as the leader while CSDs serve as the followers. Firstly, the leader provides an initial price of resources, and then followers determine the amount of resources they need to purchase. Afterwards, the leader will adjust its price accordingly. Ultimately, they will achieve a gaming equilibrium state.

Firstly, assume the leader has already provided the price p_i , the utility optimization of CSDs can be defined as

$$\begin{aligned} & \max_{q_i} U_d(q_i | p_i) \\ \text{s.t. } & \begin{cases} \alpha_i F(C_i) + \beta_i (T_i - t_i) \geq \gamma_i p_i q_i + \delta_i E_i, \\ t_i \leq T_i, \\ q_i \geq q_i^{th} \end{cases} \end{aligned} \quad (10)$$

q_i^{th} represents the minimum amount of resources that need to be purchased to meet the task's delay tolerance.

Then, after obtaining the ideal amount of resources, which is defined as q_i , the joint utility optimization problem of CSP and CSOs can be defined as

$$\begin{aligned} & \max_{p_i} \Pi_p(p_i | q_i) \\ \text{s.t. } & \begin{cases} \sum_{i=1}^M \tau_i^d p_i q_i \geq \mu E_R + \sum_{i=1}^M \tau_i^d \eta q_i, \\ p_i \leq p_i^{max} \end{cases} \end{aligned} \quad (11)$$

4) **Game Equilibrium Points:** By taking the first and second order derivatives of (10), we can obtain

$$\frac{\partial U_d(q_i | p_i)}{\partial q_i} = \frac{\beta_i C_i}{(q_i)^2} - \gamma_i p_i \quad (12)$$

$$\frac{\partial^2 U_d(q_i | p_i)}{\partial (q_i)^2} = -\frac{2\beta_i C_i}{(q_i)^3} \quad (13)$$

Because $\frac{\partial^2 U_d(q_i | p_i)}{\partial (q_i)^2} < 0$, the utility function (10) is strictly convex and has an optimal solution within its defined domain. Using q^* represents the amount of resources to be purchased, and setting $\frac{\partial U_d(q_i | p_i)}{\partial q_i} = 0$, we can obtain

$$q_i^* = \sqrt{\frac{\beta_i C_i}{\gamma_i p_i}} \quad (14)$$

After obtaining the ideal purchase quantity q_i^* , take the first derivative of (11) and then we can get $\frac{\partial \Pi_p(p_i | q_i^*)}{\partial p_i} > 0$. Therefore, the formula (11) is a monotonically increasing function within its definition domain. Because the delay of the service needs to satisfy $t_i \leq T_i$, the minimum quantity of purchased resources can be defined as

$$q_i^{th} = \frac{C_i}{T_i - T_i^p} \quad (15)$$

Combine the formulas (14) and (15), we can obtain

$$p_i^* = \frac{\beta_i (T_i - T_i^p)^2}{\gamma_i C_i} \quad (16)$$

C. Matching Stage Based on Auction Theory

We discuss the interaction between CSOs and CSP as a resource auction transaction problem. Firstly, we introduce the strategies of the auction process, and then analyze the advantages brought by this scheme theoretically.

1) **Strategies Description:** After the demands of CSDs and the prices of CSP are known, we can define the bidding situation of CSP as $\mathbf{B}_i = \{q_{i1}, q_{i2}, q_{i3}, r_i\}$ which is corresponding to $\mathbf{C}_i$. $r_i \leq p_i$ is the price proposed by CSP to CSOs. And the bid prices of CSOs corresponding to $\mathbf{Q}_j$ can be defined as $S_j = \{q_{j1}, q_{j2}, q_{j3}, r_j\}$, where $r_j \geq p_j^{min}$ is the selling price proposed by CSO o_j . We use auction theory to describe the transaction process between CSP and CSOs.

CSP's strategy: CSP can obtain CSOs' initial prices and select the appropriate candidates based on the bids. If the amount of purchased resources is the same, the lower price will lead to a higher profit for CSP. So CSP will arrange CSOs in non-descending order according to their initial bids, and select the first h CSOs to place in the candidate list which can satisfy $c_{ik} \leq \sum_{j=1}^h \tau_{ji}^o q_{ji}$, $k \in \{1, 2, 3\}$. CSP will take the bid r_{h+1} of the $(h+1)$ -th CSO as the price of its transaction.

CSOs' strategy: The higher bid is usually more advantageous for CSOs, but it may also cause themselves not to be selected. $\sum_{i=1}^M \tau_{ji}^o = 0$ represents that CSO o_j is not been selected, and will cause $\tau_j^o = 0$. $\sum_{i=1}^M \tau_{ji}^o \geq 1$ represents that CSO o_j has been selected, and will cause $\tau_j^o = 1$.

2) **Utilities of CSP and CSOs:** We define the utility of CSP as U_p , which can be calculated by the amount payments from CSDs minus the amount payments to CSOs

$$U_p = \sum_{i=1}^M \tau_i^d p_i q_i - \sum_{j=1}^N \tau_j^o (p_j q_j) \quad (17)$$

At the same time, the utility of CSOs can be defined as U_o , which mainly includes the amount payments from CSP minus the energy consumption and maintenance fee.

$$U_o = \sum_{j=1}^N \tau_j^o (p_j - \eta) q_j - \mu E_R \quad (18)$$

3) **Problem Analysis:** In this process, the prices paid by CSP can never be higher than its initial price, and the prices obtained by CSOs can never be lower than their initial bids, which will lead to a win-win situation. If CSOs reduce the bids to be selected, it will be detrimental to their income; or if they raise the bids to benefit themselves, it may also lead to the result that they are not selected. At the same time, since $\sum_{i=1}^M \tau_i^d q_i = \sum_{j=1}^N \tau_j^o q_j$ satisfies resource conservation, it can be known that $\Pi_p = U_p + U_o$. Therefore, the utility of the entire auction process is conserved, and CSP's and CSOs' utilities contribute to the overall benefit on the resource side.

IV. SIMULATION RESULT

In this section, we use Python to simulate the experiments and set $M = 200$ and $N = 50$ as participants in CPN. Since we concentrate on discussing the interaction between CSDs and CSP, without loss of generality we can suppose that every CSD requires the same amount of resources. According to [14], the maximum tolerable delay can be set as 60ms, and we suppose that the transmission delay is no more than 20ms. The weight factors are set as follows: $\alpha = 2$, $\beta = 50$, $\gamma = 0.1$, and $\delta = 0.08$. And we will randomly set the auction bids of CSOs in range of [200, 300].

As shown in Fig. 2, we use the gradient update algorithm to derive the equilibrium of the game between CSDs and CSP. It can be seen that as the number of iterations increases, the joint utility of CSP and CSOs exhibits an upward trend, ultimately converges to an optimal value, which will cause that the transaction with more CSDs results in a greater joint utility. However, when the number of iterations is small, the utility loss will also show a cumulative effect.

As shown in Fig.3, we comparatively analyze CSP's utility brought by the proposed auction scheme with the existing order-first scheme. It is evident that as the number of CSDs increases, both schemes exhibit an upward trend. The prices of CSDs matched by the order-first scheme exhibit some degree of randomness, whereas the auction scheme often prioritizes selecting from lower-priced CSOs so it can yield higher CSP's utility. Simultaneously, we set the tolerable delay to be 70ms and 80ms levels respectively, and as the tolerable delay increases the utility of CSP is decreasing.

Fig.4 illustrates the relationship between the number of CSDs and the utility of CSOs. As the number of CSDs

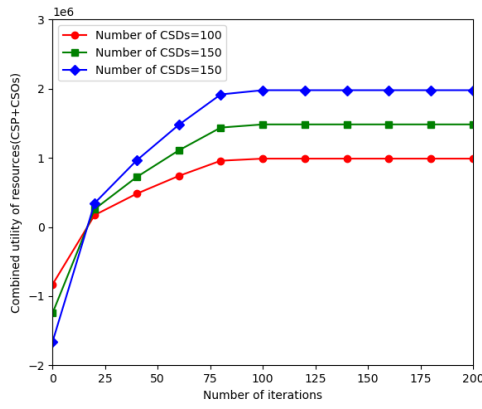


Fig. 2. Combined utility versus number of iterations

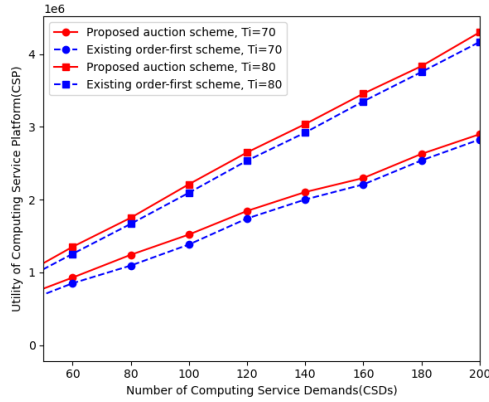


Fig. 3. Utility of CSP versus number of CSOs

increases, the utility of CSOs shows an increasing trend. The transaction with fewer CSOs can obtain higher utility when the number of CSDs is small due to the strong price competitions. However, when CSOs' resources are insufficient to meet the growing demands, the utility of CSOs will converge to a maximum value. It can be seen that the transaction with fewer CSOs will satisfy fewer demands of CPN, so that the convergence point will also be lower.

V. CONCLUSION AND FUTURE WORK

In order to enhance the enthusiasm of participants in CPN transaction, we proposed a three-tier and two-stage resource transaction mechanism, in which the participants were multiple CSOs, single CSP and multiple CSDs. The mechanism utilized Stackelberg game to describe the pricing game between CSDs and CSP, and also employed the auction theory to match suitable CSDs to optimize the resource matching issues. The simulation results showed the feasibility of our proposed mechanism. However, our work also has some limitations, such as the identity verification or requirement mapping. We will conduct further research to break through them.

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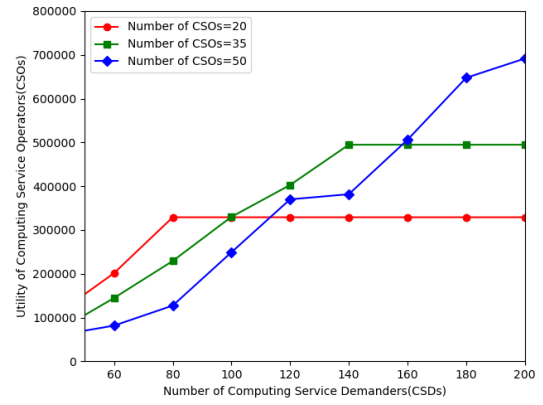


Fig. 4. Utility of CSOs versus number of CSDs

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